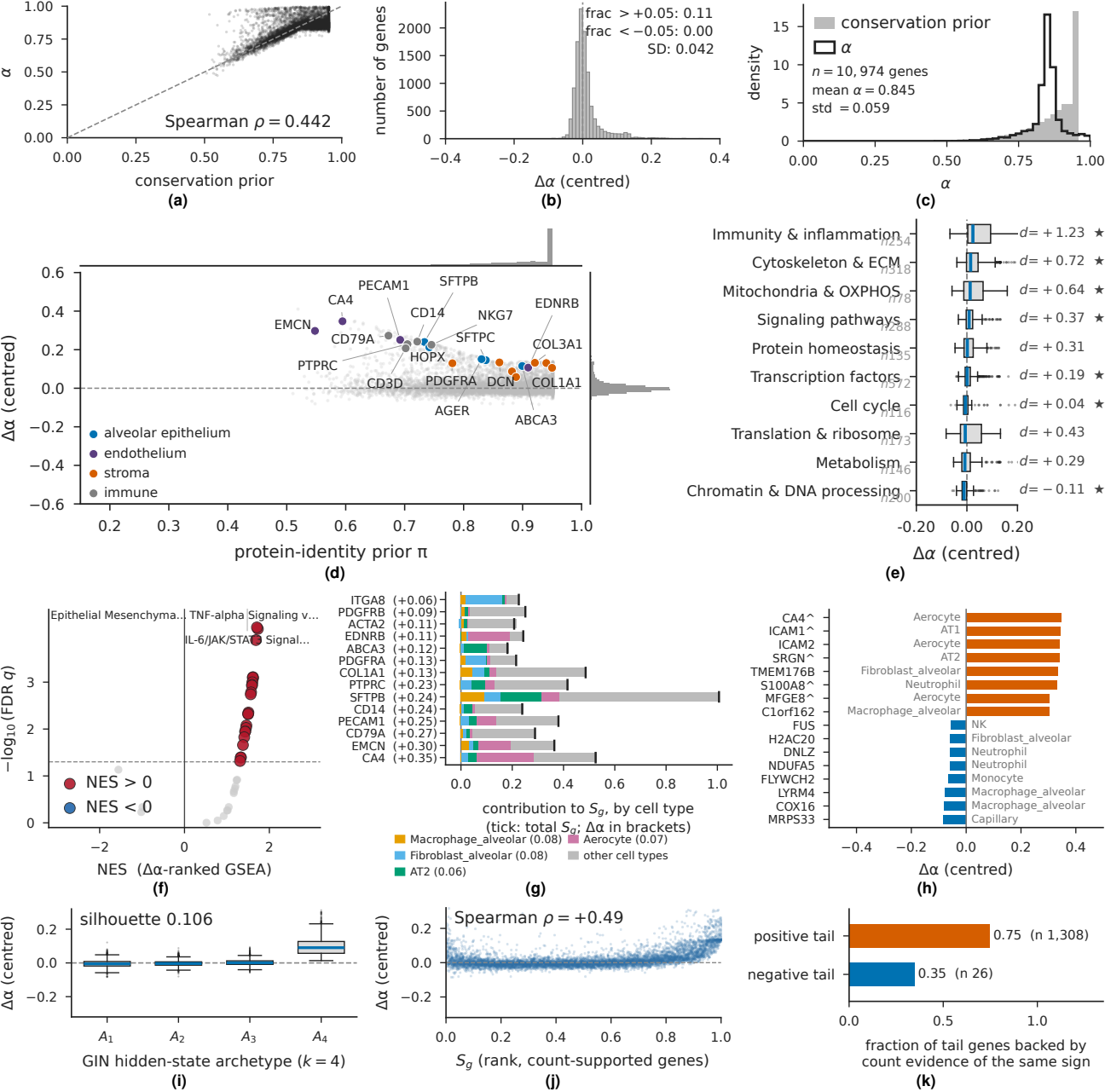

Spearman ρ = 0.442
conservation prior
α
(a)
frac > +0.05: 0.11
frac < −0.05: 0.00
SD: 0.042
number of genes
Δα (centred)
(b)
conservation prior
α
n = 10,974 genes
mean α = 0.845
std = 0.059
density
(c)
alveolar epithelium
endothelium
stroma
immune
PECAM1
SFTPB
CA4
CD14
NKG7
EDNRB
EMCN
CD79A
SFTPC
COL3A1
HOPX
PTPRC
CD3D
PDGFRA
DCN
COL1A1
AGER
ABCA3
Δα (centred)
protein-identity prior π
(d)
Immunity & inflammation n254 d = +1.23 ★
Cytoskeleton & ECM n318 d = +0.72 ★
Mitochondria & OXPHOS n78 d = +0.64 ★
Signaling pathways n288 d = +0.37 ★
Protein homeostasis n135 d = +0.31
Transcription factors n372 d = +0.19 ★
Cell cycle n116 d = +0.04 ★
Translation & ribosome n173 d = +0.43
Metabolism n146 d = +0.29
Chromatin & DNA processing n200 d = −0.11 ★
Δα (centred)
(e)
Epithelial Mesenchyma...
TNF-alpha Signaling v...
IL-6/JAK/STAT3 Signal...
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
ITGA8 (+0.06)
PDGFRB (+0.09)
ACTA2 (+0.11)
EDNRB (+0.11)
ABCA3 (+0.12)
PDGFRA (+0.13)
COL1A1 (+0.13)
PTPRC (+0.23)
SFTPB (+0.24)
CD14 (+0.24)
PECAM1 (+0.25)
CD79A (+0.27)
EMCN (+0.30)
CA4 (+0.35)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
Macrophage_alveolar (0.08)
Aerocyte (0.07)
Fibroblast_alveolar (0.08)
other cell types
AT2 (0.06)
(g)
CA4^
ICAM1^
ICAM2
SRGN^
TMEM176B
S100A8^
MFGE8^
C1orf162
FUS
H2AC20
DNLZ
NDUFA5
FLYWCH2
LYRM4
COX16
MRPS33
Aerocyte
AT1
Aerocyte
AT2
Fibroblast_alveolar
Neutrophil
Aerocyte
Macrophage_alveolar
NK
Fibroblast_alveolar
Neutrophil
Neutrophil
Monocyte
Macrophage_alveolar
Macrophage_alveolar
Capillary
Δα (centred)
(h)
silhouette 0.106
Δα (centred)
A_1
A_2
A_3
A_4
GIN hidden-state archetype ($k = 4$)
(i)
Spearman ρ = +0.49
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.75 (n 1,308)
negative tail
0.35 (n 26)
fraction of tail genes backed by count evidence of the same sign
(k)